UNIT 4

Chemical Bonding and Molecular Structure

Scientists are constantly discovering new compounds, orderly arranging the facts about them, trying to explain with the existing knowledge, organising to modify the earlier views or evolve theories for explaining the newly observed facts.

What this unit sets out to do

- understand the Kössel-Lewis approach to chemical bonding, explain the octet rule and its limitations, and draw Lewis structures of simple molecules
- explain the formation of different types of bonds, describe the VSEPR theory and predict the geometry of simple molecules
- explain the valence bond approach for the formation of covalent bonds, predict the directional properties of covalent bonds, and explain the different types of hybridisation involving s, p and d orbitals and draw shapes of simple covalent molecules
- describe the molecular orbital theory of homonuclear diatomic molecules, and explain the concept of hydrogen bond

Why atoms combine at all

- matter is made up of one or different types of element, and under normal conditions no element exists as an independent atom in nature except the noble gases, however a group of atoms is found to exist together as one species having characteristic properties, and such a group is called a molecule
- there must be some force which holds these constituent atoms together, and the attractive force which holds various constituents, atoms or ions, together in different chemical species is called a chemical bond
- since the formation of chemical compounds takes place by combination of atoms of various elements in different ways, it raises many questions, why atoms combine, why only certain combinations are possible, why some atoms combine while certain others do not, and why molecules possess definite shapes
- to answer such questions different theories have been put forward, the Kössel-Lewis approach, the Valence Shell Electron Pair Repulsion theory, the Valence Bond theory and the Molecular Orbital theory, and every system tends to be more stable, bonding being nature's way of lowering the energy of the system to attain stability

In this unit

4.1 Kössel-Lewis Approach

4.2 Ionic or Electrovalent Bond

4.3 Bond Parameters

4.4 VSEPR Theory

4.5 Valence Bond Theory

4.6 Hybridisation

4.7 Molecular Orbital Theory

4.8 Hydrogen Bonding

4.1 Kössel-Lewis approach to chemical bonding

Lewis and the cube

- in order to explain the formation of a chemical bond in terms of electrons a number of attempts were made, but it was only in 1916 that Kössel and Lewis succeeded independently in giving a satisfactory explanation, being the first to provide some logical explanation of valence based on the inertness of the noble gases
- Lewis pictured the atom in terms of a positively charged Kernel, the nucleus plus the inner electrons, and an outer shell that could accommodate a maximum of eight electrons, further assuming that these eight occupy the corners of a cube surrounding the Kernel, so the single outer shell electron of sodium would occupy one corner while in a noble gas all eight corners would be occupied, this octet representing a particularly stable arrangement
- he postulated that atoms achieve the stable octet when they are linked by chemical bonds, which in the case of sodium and chlorine happens by the transfer of an electron from sodium to chlorine giving the Na+ and Cl- ions, while in molecules like Cl2, H2 and F2 the bond is formed by the sharing of a pair of electrons between the atoms, each atom thereby attaining a stable outer octet

Lewis symbols, and what they signify

- in the formation of a molecule only the outer shell electrons take part in chemical combination and they are known as valence electrons, the inner shell electrons being well protected and generally not involved
- G.N. Lewis, an American chemist, introduced simple notations to represent valence electrons in an atom, called Lewis symbols, in which the number of dots around the symbol represents the number of valence electrons
- this number helps to calculate the common or group valence of the element, which is generally either equal to the number of dots or 8 minus the number of dots

What Kössel drew attention to

- in the periodic table the highly electronegative halogens and the highly electropositive alkali metals are separated by the noble gases
- the formation of a negative ion from a halogen atom and a positive ion from an alkali metal atom is associated with the gain and loss of an electron by the respective atoms
- the negative and positive ions thus formed attain stable noble gas electronic configurations, the noble gases, with the exception of helium which has a duplet of electrons, having a particularly stable arrangement

Electrovalence, and what Kössel's view gave us

- the electrovalence is equal to the number of unit charges on the ion, so calcium is assigned a positive electrovalence of two while chlorine has a negative electrovalence of one
- Kössel's postulations provide the basis for the modern concepts of ion-formation by electron transfer and the formation of ionic crystalline compounds, and his views have proved of great value in understanding and systematising the ionic compounds, though he did recognise that a large number of compounds did not fit into these concepts

4.1.1 Octet rule, and 4.1.2 covalent bond

The rule, and Langmuir's refinement

- Kössel and Lewis in 1916 developed the electronic theory of chemical bonding, according to which atoms can combine either by transfer of valence electrons from one atom to another or by sharing of valence electrons in order to have an octet in their valence shells, and this is known as the octet rule
- Langmuir in 1919 refined the Lewis postulations by abandoning the idea of the stationary cubical arrangement of the octet and by introducing the term covalent bond
- the chlorine atom, with the configuration neon core plus 3s2 3p5, is one electron short of the argon configuration, and the formation of Cl2 can be understood as the sharing of a pair of electrons between the two chlorine atoms, each contributing one electron to the shared pair, so that both attain the outer shell octet of argon, the dots in such a diagram representing electrons and the diagrams being called Lewis dot structures

The conditions, and multiple bonds

- each bond is formed as a result of sharing an electron pair between the atoms, each combining atom contributes at least one electron to the shared pair, and the combining atoms attain the outer-shell noble gas configurations as a result of the sharing
- when two atoms share one electron pair they are said to be joined by a single covalent bond, and in many compounds there are multiple bonds, the formation of which envisages sharing of more than one electron pair
- if two atoms share two pairs of electrons the covalent bond between them is called a double bond, for example the carbon dioxide molecule has two double bonds between the carbon and oxygen atoms, and in ethene the two carbon atoms are joined by a double bond

Molecules and ions drawn with every valence electron shown as a dot

Shared pairs sit between the bonded atoms, lone pairs on the atom that owns them

Each H atom attains the configuration of helium, a duplet of electrons

Figure 4.A The Lewis Representation of Some Molecules

Triple bonds

- when combining atoms share three electron pairs, as with the two nitrogen atoms in N2 and the two carbon atoms in ethyne, a triple bond is formed

4.1.3 Lewis representation of simple molecules

The steps for writing one

- Lewis dot structures provide a picture of bonding in molecules and ions in terms of shared pairs of electrons and the octet rule, and while such a picture may not explain the bonding and behaviour of a molecule completely, it does help in understanding its formation and properties to a large extent
- the total number of electrons required is obtained by adding the valence electrons of the combining atoms, so in CH4 there are eight valence electrons available for bonding, four from carbon and four from the four hydrogen atoms
- for anions each negative charge means the addition of one electron and for cations each positive charge means subtracting one, so the carbonate ion has two additional electrons beyond those of the neutral atoms while the ammonium ion has lost one
- knowing the chemical symbols and the skeletal structure, it is easy to distribute the total electrons as bonding shared pairs in proportion to the total bonds, the least electronegative atom generally occupying the central position, so in NF3 and the carbonate ion nitrogen and carbon are central while fluorine and oxygen occupy the terminal positions, and after accounting for the single bonds the remaining pairs are either used for multiple bonding or remain as lone pairs, the basic requirement being that each bonded atom gets an octet

Problem 4.1, the carbon monoxide molecule

- step 1 counts the total valence electrons of carbon and oxygen, whose outer configurations are 2s2 2p2 and 2s2 2p4, giving 4 plus 6, that is 10
- step 2 writes the skeletal structure as C next to O, and step 3 draws a single bond between them and completes the octet on O, leaving the remaining two electrons as a lone pair on C
- that does not complete the octet on carbon, so we have to resort to multiple bonding, in this case a triple bond, between C and O, which satisfies the octet rule for both atoms

Problem 4.2, the nitrite ion

- step 1 counts the valence electrons of the nitrogen atom, the two oxygen atoms and the additional negative charge, which is worth one electron, giving 5 plus twice 6 plus 1, that is 18 electrons
- step 2 writes the skeletal structure with nitrogen between the two oxygens, and step 3 draws a single bond from nitrogen to each oxygen and completes the octets on the oxygens, which does not complete the octet on nitrogen if the remaining two electrons are a lone pair on it
- so we have to resort to multiple bonding between nitrogen and one of the oxygen atoms, in this case a double bond

4.1.4 Formal charge

What a formal charge is

- Lewis dot structures in general do not represent the actual shapes of molecules, and in a polyatomic ion the net charge is possessed by the ion as a whole and not by a particular atom, though it is feasible to assign a formal charge to each atom
- the formal charge of an atom in a polyatomic molecule or ion is the difference between the number of valence electrons of that atom in an isolated or free state and the number of electrons assigned to it in the Lewis structure, that is the valence electrons in the free atom minus the lone pair electrons minus half the shared electrons

Formal charges in ozone

- the counting is based on the assumption that the atom in the molecule owns one electron of each shared pair and both the electrons of a lone pair
- numbering the three oxygen atoms 1, 2 and 3, the central atom has 6 minus 2 minus half of 6, that is +1, the end atom marked 2 has 6 minus 4 minus half of 4, that is 0, and the end atom marked 3 has 6 minus 6 minus half of 2, that is -1
- formal charges do not indicate real charge separation within the molecule, and indicating them only helps in keeping track of the valence electrons, but they do help in selecting the lowest energy structure from a number of possible Lewis structures, generally the one with the smallest formal charges on the atoms, the formal charge being a factor based on a pure covalent view of bonding in which electron pairs are shared equally by neighbouring atoms

4.1.5 Limitations of the octet rule

The incomplete octet, and odd-electron molecules

- the octet rule, though useful, is not universal, being quite useful for understanding the structures of most organic compounds and applying mainly to the second period elements, and there are three types of exception to it
- in some compounds the number of electrons surrounding the central atom is less than eight, especially with elements having less than four valence electrons, examples being LiCl, BeH2 and BCl3, since Li, Be and B have only 1, 2 and 3 valence electrons, and other such compounds are AlCl3 and BF3
- in molecules with an odd number of electrons like nitric oxide and nitrogen dioxide, the octet rule is not satisfied for all the atoms

The expanded octet

- elements in and beyond the third period of the periodic table have, apart from 3s and 3p orbitals, 3d orbitals also available for bonding

The rest of the octet's troubles

- sulphur also forms many compounds in which the octet rule is obeyed, so in sulphur dichloride the S atom has an octet of electrons around it
- the octet rule is based upon the chemical inertness of the noble gases, however some noble gases such as xenon and krypton also combine with oxygen and fluorine to form compounds like XeF2, KrF2 and XeOF2
- the theory does not account for the shape of molecules, and it does not explain the relative stability of molecules, being totally silent about the energy of a molecule

4.2 Ionic or electrovalent bond

What ionic bond formation depends on

- the formation of ionic compounds depends primarily upon the ease of formation of the positive and negative ions from the respective neutral atoms, and upon the arrangement of those ions in the solid, that is the lattice of the crystalline compound
- the formation of a positive ion involves ionization, the removal of electrons from the neutral atom, and that of a negative ion involves the addition of electrons, the electron gain enthalpy being the enthalpy change when a gas phase atom in its ground state gains an electron, a process that may be exothermic or endothermic, while ionization is always endothermic, and electron affinity is the negative of the energy change accompanying electron gain
- so ionic bonds will be formed more easily between elements with comparatively low ionization enthalpies and elements with comparatively high negative values of electron gain enthalpy, and most ionic compounds have cations derived from metallic elements and anions from non-metallic ones, the ammonium ion, made of two non-metallic elements, being an exception

Why the lattice is what stabilises the compound

- ionic compounds in the crystalline state consist of orderly three-dimensional arrangements of cations and anions held together by coulombic interaction energies, crystallising in different structures determined by the size of the ions, their packing arrangements and other factors, the rock salt structure of sodium chloride being the standard example
- in ionic solids the sum of the electron gain enthalpy and the ionization enthalpy may be positive but the crystal still gets stabilized by the energy released in forming the lattice, so the ionization enthalpy for Na+ formation from Na is 495.8 kJ mol-1 while the electron gain enthalpy for chlorine is only -348.7 kJ mol-1, and their sum of 147.1 kJ mol-1 is more than compensated for by the enthalpy of lattice formation of solid NaCl at -788 kJ mol-1

Two bonded atoms A and B, with the bond length R measured centre to centre

R is the sum of the two covalent radii, rA and rB

Figure 4.B The bond length in a covalent molecule AB. R = rA + rB (R is the bond length and rA and rB are the covalent radii of atoms A and B respectively)

Two chlorine atoms drawn twice over, as small inner circles and larger outer ones

Half the 198 pm between bonded atoms gives the covalent radius, 99 pm

Half the 360 pm between atoms of separate molecules gives the van der Waals radius, 180 pm

Figure 4.C Covalent and van der Waals radii in a chlorine molecule. The inner circles correspond to the size of the chlorine atom (rvdw and rc are van der Waals and covalent radii respectively).

4.2.1 Lattice enthalpy, and 4.3 bond parameters

Lattice enthalpy

- a qualitative measure of the stability of an ionic compound is provided by its enthalpy of lattice formation, and not simply by achieving an octet of electrons around the ionic species in the gaseous state
- the lattice enthalpy of an ionic solid is the energy required to completely separate one mole of a solid ionic compound into gaseous constituent ions, so the lattice enthalpy of NaCl being 788 kJ mol-1 means 788 kJ is required to separate one mole of solid NaCl into one mole each of Na+ and Cl- gas at an infinite distance
- the process involves both the attractive forces between ions of opposite charge and the repulsive forces between ions of like charge, and since the solid crystal is three-dimensional it is not possible to calculate the lattice enthalpy from those forces alone, factors associated with the crystal geometry having to be included

Covalent radius and van der Waals radius

- in a covalent bond the contribution from each atom is called the covalent radius of that atom, measured approximately as the radius of an atom's core which is in contact with the core of an adjacent atom in a bonded situation, and it is half the distance between two similar atoms joined by a covalent bond in the same molecule
- the van der Waals radius represents the overall size of the atom including its valence shell in a nonbonded situation, and is half the distance between two similar atoms in separate molecules in a solid

Covalent radii for the common elements, in picometres

The values cited are for single bonds, except where otherwise indicated in parenthesis

They follow the periodic trends of Unit 3

Figure 4.D Covalent Radii, *rcov/(pm)

4.3.2 Bond angle, and 4.3.3 bond enthalpy

Bond angle

- the bond angle is the angle between the orbitals containing bonding electron pairs around the central atom in a molecule or complex ion, expressed in degrees and experimentally determinable by spectroscopic methods
- it gives some idea of the distribution of orbitals around the central atom and hence helps in determining the shape, for example the H-O-H bond angle in water

Bond enthalpy

- bond enthalpy is the amount of energy required to break one mole of bonds of a particular type between two atoms in a gaseous state, its unit being kJ mol-1, so the H-H bond enthalpy in the hydrogen molecule is 435.8 kJ mol-1
- for molecules containing multiple bonds the values are higher, oxygen at 498 kJ mol-1 and nitrogen at 946.0 kJ mol-1, and the larger the bond dissociation enthalpy the stronger the bond in the molecule, while for a heteronuclear diatomic molecule like HCl it is 431.0 kJ mol-1
- in polyatomic molecules the measurement is more complicated, for example in water the enthalpy needed to break the two O-H bonds is not the same

Table 4.A Average Bond Lengths for Some Single, Double and Triple Bonds

Bond Type	Covalent Bond Length (pm)	Bond Type	Covalent Bond Length (pm)
O–H	96	C=O	121
C–H	107	N=O	122
N–O	136	C=C	133
C–O	143	C=N	138
C–N	143	C≡N	116
C–C	154	C≡C	120

Table 4.B Bond Lengths in Some Common Molecules

Molecule	Bond Length (pm)	Molecule	Bond Length (pm)
H2 (H – H)	74	O2 (O = O)	121
F2 (F – F)	144	HF (H – F)	92
Cl2 (Cl – Cl)	199	HCl (H – Cl)	127
Br2 (Br – Br)	228	HBr (H – Br)	141
I2 (I – I)	267	HI (H – I)	160
N2 (N ≡ N)	109		

Two equivalent structures for ozone, differing only in which bond is double

Neither one alone matches what is measured, since both bonds are the same length

The real molecule is a single hybrid lying between the two canonical forms

Figure 4.E Resonance in the O3 molecule

Mean bond enthalpy

- breaking the first O-H bond of water needs 502 kJ mol-1 and the second 427 kJ mol-1, and the difference shows that the second bond undergoes some change because of the changed chemical environment, which is also why the same O-H bond differs in energy between ethanol and water
- so in polyatomic molecules the term mean or average bond enthalpy is used, obtained by dividing the total bond dissociation enthalpy by the number of bonds broken, which for water is 502 plus 427 over 2, that is 464.5 kJ mol-1

4.3.4 Bond order, and 4.3.5 resonance structures

Bond order

- in the Lewis description of a covalent bond, the bond order is the number of bonds between the two atoms in a molecule, so it is 1 in H2 with a single shared pair, 2 in O2 with two shared pairs and 3 in N2 with three, and 3 in CO
- for N2 the bond order is 3 and its bond enthalpy is 946 kJ mol-1, one of the highest for a diatomic molecule, while isoelectronic molecules and ions have identical bond orders, so F2 and the peroxide ion both have bond order 1 and N2, CO and NO+ all have bond order 3
- a general correlation useful for understanding the stabilities of molecules is that with increase in bond order the bond enthalpy increases and the bond length decreases

Why one Lewis structure is sometimes not enough

- it is often observed that a single Lewis structure is inadequate for the representation of a molecule in conformity with its experimentally determined parameters
- the ozone molecule, for example, can be equally represented by either of two structures, which are called its canonical forms, the real molecule being the resonance hybrid of them

Three equivalent structures for the carbonate ion, the double bond moving between the three oxygens

All three carbon to oxygen bonds are found to be equivalent

The ion is best described as the hybrid of the three

Figure 4.F Resonance in CO32–, I, II and III represent the three canonical forms.

Three structures for carbon dioxide, differing in where the double and triple bonds sit

The measured C to O length, 115 pm, lies between the double bond value of 121 pm and the triple bond value of 110 pm

No single structure can show that, so the molecule is the hybrid of the three

Figure 4.G Resonance in CO2 molecule, I, II and III represent the three canonical forms.

Problems 4.3 and 4.4, and what resonance does

- Problem 4.3 explains the carbonate ion in terms of resonance, since a single Lewis structure with two single bonds and one double bond represents unequal bonds, whereas experimentally all carbon to oxygen bonds in the ion are equivalent, so it is best described as a resonance hybrid of three canonical forms
- Problem 4.4 explains carbon dioxide, whose measured carbon to oxygen bond length of 115 pm lies between the values for a normal double bond at 121 pm and a triple bond at 110 pm, which no single Lewis structure can depict, so the molecule is best described as a hybrid of three canonical forms
- in general, resonance stabilizes the molecule because the energy of the hybrid is less than that of any single canonical structure, and resonance averages the bond characteristics as a whole, so the energy of the ozone hybrid is lower than either of its two canonical forms

The misconceptions to dispel

- the canonical forms have no real existence, and the molecule does not exist for a certain fraction of time in one form and for other fractions in others
- there is no such equilibrium between the canonical forms as there is between tautomeric forms, the keto and enol forms, in tautomerism
- the molecule as such has a single structure, which is the resonance hybrid of the canonical forms and which cannot be depicted by a single Lewis structure

4.3.6 Polarity of bonds

Polar and nonpolar covalent bonds

- the existence of a hundred percent ionic or covalent bond represents an ideal situation, and in reality no bond or compound is either completely covalent or ionic, even the covalent bond between two hydrogen atoms having some ionic character
- when a covalent bond is formed between two similar atoms, as in H2, O2, Cl2, N2 or F2, the shared pair is equally attracted by the two atoms, so it sits exactly between the two identical nuclei and the bond is called a nonpolar covalent bond
- in a heteronuclear molecule like HF the shared pair is displaced more towards fluorine, since the electronegativity of fluorine is far greater than that of hydrogen, and the resultant bond is a polar covalent bond

What a dipole moment is

- as a result of polarisation the molecule possesses a dipole moment, defined as the product of the magnitude of the charge and the distance between the centres of positive and negative charge, usually designated by the Greek letter μ, so that μ is charge times distance of separation
- dipole moment is usually expressed in Debye units, the conversion factor being 1 D = 3.33564 × 10-30 C m, where C is coulomb and m is metre
- it is a vector quantity, by convention depicted by a small arrow with its tail on the negative centre and head pointing towards the positive centre, though in chemistry the presence of a dipole moment is shown by a crossed arrow on the Lewis structure, the cross on the positive end and the arrow head on the negative, symbolising the direction of the shift of electron density, which is opposite to the conventional dipole moment vector
- Peter Debye, the Dutch chemist, received the Nobel prize in 1936 for his work on X-ray diffraction and dipole moments, and the magnitude of the dipole moment is given in Debye units in order to honour him

Adding bond dipoles up

- in polyatomic molecules the dipole moment depends not only on the individual bond dipoles but also on the spatial arrangement of the bonds, so the dipole moment of a molecule is the vector sum of the dipole moments of its bonds
- in the water molecule, which has a bent structure, the two O-H bonds are oriented at an angle of 104.5 degrees, and the net dipole moment of 6.17 × 10-30 C m, that is 1.85 D, is the resultant of the two bond dipoles
- the dipole moment of BeF2 is zero, because the two equal bond dipoles point in opposite directions and cancel each other, and in the tetra-atomic BF3 it is also zero although the B-F bonds are oriented at 120 degrees to one another, since the resultant of any two is equal and opposite to the third

The case of ammonia against nitrogen trifluoride

- both molecules have a pyramidal shape with a lone pair of electrons on the nitrogen atom, and although fluorine is more electronegative than hydrogen, the two molecules do not behave alike

Why ammonia's dipole beats that of nitrogen trifluoride

- the dipole moment of NH3 at 4.90 × 10-30 C m is greater than that of NF3 at 0.8 × 10-30 C m
- in NH3 the orbital dipole due to the lone pair is in the same direction as the resultant dipole moment of the N-H bonds, whereas in NF3 the orbital dipole is in the direction opposite to the resultant of the three N-F bond moments
- so the orbital dipole of the lone pair decreases the effect of the resultant N-F bond moments, which is what gives NF3 its low dipole moment

Fajans' rules on partial covalent character

- just as all covalent bonds have some partial ionic character, ionic bonds also have partial covalent character, which Fajans discussed in terms of a set of rules
- the smaller the size of the cation and the larger the size of the anion, the greater the covalent character of an ionic bond, and the greater the charge on the cation, the greater the covalent character
- for cations of the same size and charge, the one with the configuration n minus 1 d to the n ns0, typical of transition metals, is more polarising than the one with a noble gas configuration ns2 np6, typical of alkali and alkaline earth metal cations
- the cation polarises the anion, pulling the electronic charge towards itself and thereby increasing the electronic charge between the two, which is precisely what happens in a covalent bond, so the polarising power of the cation, the polarisability of the anion and the extent of distortion of the anion are the factors that determine the per cent covalent character of the ionic bond

4.4 The valence shell electron pair repulsion theory

What VSEPR is for

- the Lewis concept is unable to explain the shapes of molecules, and this theory provides a simple procedure to predict the shapes of covalent molecules

Table 4.C Dipole Moments of Selected Molecules

Type of Molecule	Example	Dipole Moment, μ(D)	Geometry
Molecule (AB)	HF	1.78	linear
	HCl	1.07	linear
	HBr	0.79	linear
	Hl	0.38	linear
	H2	0	linear
Molecule (AB2)	H2O	1.85	bent
	H2S	0.95	bent
	CO2	0	linear
Molecule (AB3)	NH3	1.47	trigonal-pyramidal
	NF3	0.23	trigonal-pyramidal
	BF3	0	trigonal-planar
Molecule (AB4)	CH4	0	tetrahedral
	CHCl3	1.04	tetrahedral
	CCl4	0	tetrahedral

Ball and stick models for BF3, CH4 and PCl5

Trigonal planar, tetrahedral and trigonal-bipyramidal in turn

Every pair around the central atom is a bonding pair

Figure 4.H The shapes of molecules in which central atom has no lone pair

The postulates

- Sidgwick and Powell in 1940 proposed a simple theory based on the repulsive interactions of the electron pairs in the valence shell of the atoms, further developed and redefined by Nyholm and Gillespie in 1957
- the shape of a molecule depends upon the number of valence shell electron pairs, bonded or nonbonded, around the central atom, and pairs of electrons in the valence shell repel one another since their electron clouds are negatively charged, so they tend to occupy positions that minimise repulsion and maximise the distance between them
- the valence shell is taken as a sphere with the electron pairs localising on the spherical surface at maximum distance from one another, a multiple bond is treated as if it were a single electron pair, its two or three pairs treated as a single super pair, and where two or more resonance structures can represent a molecule the model applies to any of them

Lone pairs push harder

- the repulsive interaction of electron pairs decreases in the order lone pair against lone pair, then lone pair against bond pair, then bond pair against bond pair
- Nyholm and Gillespie refined the model by explaining the important difference between lone pairs and bonding pairs, since lone pairs are localised on the central atom while each bonded pair is shared between two atoms, so lone pair electrons occupy more space and repel more strongly, and these repulsion effects result in deviations from idealised shapes and alterations in bond angles
- for predicting geometrical shapes it is convenient to divide molecules into two categories, those in which the central atom has no lone pair and those in which it has one or more

What the theory gets right

- in the compounds AB2, AB3, AB4, AB5 and AB6 the arrangement of electron pairs and the B atoms around the central atom A are linear, trigonal planar, tetrahedral, trigonal-bipyramidal and octahedral respectively
- the theory predicts the geometry of a large number of molecules, especially the compounds of p-block elements, accurately, and is also quite successful even when the energy difference between possible structures is very small

Each row a number of bonding pairs, from two up to six	Beside it the arrangement those pairs take and the geometry of the molecule
Linear, trigonal planar, tetrahedral, trigonal-bipyramidal and octahedral in turn, with an example of each	

Figure 4.I Geometry of Molecules in which the Central Atom has No Lone Pair of Electrons

What this page carries

- the whole of this page is given over to the table of geometries for molecules whose central atom carries only bonding pairs
- the table is printed as a drawing rather than as text, so it is described here rather than reproduced

Each row a count of bonding pairs and of lone pairs on the central atom

Beside it the arrangement of all the pairs, and then the shape the atoms alone make

Bent, trigonal pyramidal, see-saw, T-shaped and square planar among them

Figure 4.J Shape (geometry) of Some Simple Molecules/Ions with Central Ions having One or More Lone Pairs of Electrons(E).

What this page carries

- the whole of this page is given over to the table of shapes for molecules and ions whose central atom carries one or more lone pairs
- the lone pairs are counted in the arrangement of electron pairs but not in the shape, which names only where the atoms sit
- the table is printed as a drawing rather than as text, so it is described here rather than reproduced

Molecule type, then the count of bonding and of lone pairs, then the arrangement

Then the shape, and beside it the reason that shape is acquired

Each reason turns on lone pair repulsion being stronger than bond pair repulsion

Figure 4.K Shapes of Molecules containing Bond Pair and Lone Pair

What this page carries

- the whole of this page is given over to the table that explains the reasons for the distortions in the geometry of a molecule
- theoretically the shape of SO2 should have been triangular planar, but the lone pair on sulphur pushes the two bonds together, which is the kind of reasoning the table sets out for each case
- the table is printed as a drawing rather than as text, so it is described here rather than reproduced

The last row of the distortion table

- for a molecule of type AB3E2, three bonding pairs and two lone pairs, the shape is T-shaped
- the reason is that with the lone pairs at equatorial positions there are fewer lone pair-bond pair repulsions than in the arrangements where the lone pairs are axial, so that structure is the most stable

4.5 Valence bond theory

Why another theory was needed

- the Lewis approach helps in writing the structure of molecules but fails to explain the formation of a chemical bond, and gives no reason for the difference in bond dissociation enthalpies and bond lengths in molecules like H2 at 435.8 kJ mol-1 and 74 pm and F2 at 155 kJ mol-1 and 144 pm, although in both cases a single covalent bond is formed by sharing an electron pair, and it gives no idea about the shapes of polyatomic molecules
- the VSEPR theory gives the geometry of simple molecules but does not explain it theoretically and has limited applications, so to overcome these limitations two theories based on quantum mechanical principles are introduced, valence bond theory and molecular orbital theory

Two hydrogen atoms approaching

- valence bond theory was introduced by Heitler and London in 1927 and developed further by Pauling and others, resting on the knowledge of atomic orbitals, electronic configurations, the overlap criteria of atomic orbitals, hybridization and the principles of variation and superposition, and it is treated here qualitatively and without mathematics
- consider two hydrogen atoms A and B approaching each other, with nuclei NA and NB and electrons eA and eB, so that when the two atoms are at a large distance there is no interaction between them, but as they approach, new attractive and repulsive forces begin to operate
- attractive forces arise between the nucleus of one atom and its own electron, and between the nucleus of one atom and the electron of the other

Two approaching hydrogen atoms with the forces drawn on them

Attraction between each nucleus and both electrons

Repulsion between the two electrons and between the two nuclei

Figure 4.L Forces of attraction and repulsion during the formation of H2 molecule

Potential energy plotted against the distance between the two nuclei

The curve falls to a minimum and then rises steeply as the nuclei are pushed closer

The minimum sits at 74 pm and is the most stable state of the molecule

Figure 4.M The potential energy curve for the formation of H2 molecule as a function of internuclear distance of the H atoms. The minimum in the curve corresponds to the most stable state of H2.

Which forces win, and what the minimum means

- repulsive forces arise between the electrons of the two atoms and between the two nuclei, so attractive forces tend to bring the atoms close together whereas repulsive forces tend to push them apart
- experimentally the magnitude of the new attractive force is more than the new repulsive forces, so the two atoms approach and the potential energy decreases, until a stage is reached where the net attraction balances the repulsion and the system acquires minimum energy, at which point the two hydrogen atoms are bonded with a bond length of 74 pm
- since energy is released when the bond forms, the hydrogen molecule is more stable than the isolated atoms, and the energy released is the bond enthalpy, corresponding to the minimum in the curve, so conversely 435.8 kJ of energy is required to dissociate one mole of H2

4.5.1 Orbital overlap concept

What overlapping means

- in the formation of the hydrogen molecule there is a minimum energy state when two hydrogen atoms are so near that their atomic orbitals undergo partial interpenetration, and this partial merging is called the overlapping of atomic orbitals, which results in the pairing of electrons
- the extent of overlap decides the strength of a covalent bond, and in general the greater the overlap the stronger the bond, so according to the orbital overlap concept the formation of a covalent bond between two atoms results by pairing of electrons present in the valence shell having opposite spins

Pairs of orbitals brought together in different relative phases and directions

Same phase facing each other gives positive overlap, opposite phase gives negative

Perpendicular approach gives zero overlap, the signs marking phase and not charge

Figure 4.N Positive, negative and zero overlaps of s and p atomic orbitals

4.5.2 Directional properties of bonds, and 4.5.3 overlapping of atomic orbitals

Why direction matters

- the covalent bond is formed by overlapping of atomic orbitals, the hydrogen molecule forming by overlap of the 1s orbitals of two H atoms, but in polyatomic molecules like CH4, NH3 and H2O the geometry matters as well as the bond formation, so we must ask why CH4 is tetrahedral with HCH angles of 109.5 degrees and why NH3 is pyramidal
- when the orbitals of two atoms come close to form a bond their overlap may be positive, negative or zero, depending on the sign or phase and the direction of orientation of the amplitude of the orbital wave function in space, the signs on the boundary surface diagrams showing phase and not charge, and orbitals forming a bond should have the same phase and orientation, which is called positive overlap

Methane, and where simple overlap fails

- the electronic configuration of carbon in its ground state is helium core plus 2s2 2p2, which in the excited state becomes helium core plus 2s1 2px1 2py1 2pz1, the energy required for this excitation being compensated by the energy released on overlap with hydrogen
- the four atomic orbitals of carbon, each with an unpaired electron, can overlap with the singly occupied 1s orbitals of four H atoms, giving four C-H bonds, but since the three p orbitals of carbon are at 90 degrees to one another the HCH angle for those would also be 90 degrees, while the 2s orbital of carbon and the 1s orbital of H are spherically symmetrical and can overlap in any direction, so the direction of the fourth C-H bond cannot be ascertained
- that description does not fit the tetrahedral HCH angles of 109.5 degrees, so simple atomic orbital overlap does not account for the directional characteristics of the bonds in CH4, and by the same argument the HNH and HOH angles would come out at 90 degrees, in disagreement with the actual 107 and 104.5 degrees

4.5.4 Types of overlapping and nature of covalent bonds

Sigma and pi

- the covalent bond may be classified into two types depending upon the type of overlapping, the sigma bond and the pi bond
- a sigma bond is formed by the end to end or head-on overlap of bonding orbitals along the internuclear axis, also called axial overlap, which can arise from s with s, from s with p, or from p with p overlapping
- in the formation of a pi bond the atomic orbitals overlap in such a way that their axes remain parallel to each other and perpendicular to the internuclear axis, the orbitals formed by this sidewise overlapping consisting of two saucer type charged clouds above and below the plane of the participating atoms

4.5.5 Strength of sigma and pi bonds

Which is stronger

- basically the strength of a bond depends upon the extent of overlapping, and in a sigma bond the overlapping takes place to a larger extent, so it is stronger than a pi bond where the overlap is smaller
- in the formation of multiple bonds between two atoms of a molecule, pi bonds are formed in addition to a sigma bond

4.6 Hybridisation

What hybridisation is

- in order to explain the characteristic geometrical shapes of polyatomic molecules like CH4, NH3 and H2O, Pauling introduced the concept of hybridisation, according to which atomic orbitals combine to form a new set of equivalent orbitals known as hybrid orbitals, and unlike pure orbitals it is the hybrid orbitals that are used in bond formation
- the phenomenon can be defined as the process of intermixing of orbitals of slightly different energies so as to redistribute their energies, resulting in a new set of orbitals of equivalent energies and shape, so when one 2s and three 2p orbitals of carbon hybridise, four new sp3 hybrid orbitals are formed
- the number of hybrid orbitals is equal to the number of atomic orbitals that get hybridised, and the hybridised orbitals are always equivalent in energy and shape

One s and one p orbital merging into two equivalent sp hybrids

The two hybrids pointing opposite ways along one axis, 180 degrees apart

Each overlapping a chlorine 2p orbital end on, giving the linear molecule

Figure 4.O (a) Formation of sp hybrids from s and p orbitals; (b) Formation of the linear BeCl2 molecule

One s and two p orbitals merging into three equivalent sp2 hybrids

The three lying flat in one plane at 120 degrees to each other

Each overlapping a chlorine 2p orbital, giving the trigonal planar molecule

Figure 4.P Formation of sp2 hybrids and the BCl3 molecule

The rest of the features, and the conditions

- hybrid orbitals are more effective in forming stable bonds than pure atomic orbitals, and they are directed in space in some preferred direction to have minimum repulsion between electron pairs and thus a stable arrangement, so the type of hybridisation indicates the geometry of the molecule
- the orbitals hybridised are those present in the valence shell of the atom, and they should have almost equal energy, while promotion of an electron is not an essential condition prior to hybridisation, and it is not necessary that only half filled orbitals participate, since in some cases even filled orbitals of the valence shell take part

4.6.1 Types of hybridisation

sp, and sp2

- sp hybridisation involves the mixing of one s and one p orbital giving two equivalent sp hybrid orbitals, each with 50 per cent s-character and 50 per cent p-character, and a molecule whose central atom is sp-hybridised and linked directly to two others has linear geometry, this type also being known as diagonal hybridisation, the two hybrids pointing in opposite directions with projecting positive lobes and very small negative lobes
- in BeCl2 the ground state configuration of Be is 1s2 2s2, and in the excited state one 2s-electron is promoted to a vacant 2p orbital to account for its bivalency, one 2s and one 2p orbital then hybridising to two sp orbitals oriented at 180 degrees, each overlapping the 2p orbital of a chlorine axially to form a Be-Cl sigma bond
- sp2 hybridisation involves one s and two p orbitals forming three equivalent sp2 orbitals, so in BCl3 the ground state configuration of boron is 1s2 2s2 2p1 and in the excited state one 2s electron is promoted to a vacant 2p orbital, leaving boron with three unpaired electrons

One s and three p orbitals of carbon merging into four equivalent sp3 hybrids

The four pointing to the corners of a tetrahedron, 109.5 degrees apart

Each overlapping a hydrogen 1s orbital, giving methane

Figure 4.Q Formation of sp3 hybrids by the combination of s, px , py and pz atomic orbitals of carbon and the formation of CH4 molecule

Three of nitrogen's four sp3 hybrids holding one electron each, the fourth a lone pair

The three bonding hybrids overlapping hydrogen 1s orbitals

The lone pair pressing the bonds together, closing the angle to 107 degrees

Figure 4.R Formation of NH3 molecule

Two of oxygen's four sp3 hybrids holding one electron each, the other two lone pairs

Two corners of the tetrahedron taken by hydrogen atoms and two by lone pairs

The angle closing to 104.5 degrees, giving the V-shape

Figure 4.S Formation of H2O molecule

Boron trichloride, and then methane

- the three orbitals of boron, one 2s and two 2p, hybridise to three sp2 hybrid orbitals oriented in a trigonal planar arrangement, overlapping the 2p orbitals of chlorine to form three B-Cl bonds, so the geometry of BCl3 is trigonal planar with a ClBCl bond angle of 120 degrees
- sp3 hybridisation is explained by methane, in which one s orbital and three p orbitals of the valence shell mix to form four sp3 hybrid orbitals of equivalent energy and shape, each with 25 per cent s-character and 75 per cent p-character, directed towards the four corners of a tetrahedron with an angle of 109.5 degrees between them

Ammonia and water by the same route

- in NH3 the valence shell configuration of nitrogen in the ground state is 2s2 2px1 2py1 2pz1, so three of the sp3 hybrid orbitals hold one unpaired electron each and a lone pair sits in the fourth, the three bonding hybrids overlapping the 1s orbitals of hydrogen to form three N-H sigma bonds
- since the repulsion between a lone pair and a bond pair is greater than that between two bond pairs, the molecule gets distorted and the bond angle is reduced to 107 degrees from 109.5, so the geometry is pyramidal
- in H2O the four oxygen orbitals, one 2s and three 2p, undergo sp3 hybridisation, two of the resulting hybrids holding one electron each and the other two a pair each, so the tetrahedron has two corners occupied by hydrogen atoms and two by lone pairs, the bond angle being reduced to 104.5 degrees and the molecule acquiring a V-shape or angular geometry

Two sp2 carbons joined head on, giving the sigma bond of the double bond

The leftover p orbital on each overlapping sidewise above and below the plane

That sidewise overlap being the pi bond, two clouds one above and one below

Figure 4.T Formation of sigma and pi bonds in ethene

4.6.2 Other examples of sp3, sp2 and sp hybridisation

sp3 in ethane

- in the ethane molecule both carbon atoms assume the sp3 hybrid state, one of the four sp3 hybrid orbitals of each carbon overlapping axially with a similar orbital of the other to form an sp3-sp3 sigma bond, while the other three hybrid orbitals of each carbon form sp3-s sigma bonds with hydrogen atoms
- so in ethane the C-C bond length is 154 pm and each C-H bond length is 109 pm

sp2 in ethene

- in the formation of ethene one sp2 hybrid orbital of a carbon atom overlaps axially with an sp2 orbital of the other carbon to form the C-C sigma bond, while the other two sp2 orbitals of each carbon make sp2-s sigma bonds with two hydrogen atoms
- the unhybridised orbital, 2px or 2py, of one carbon overlaps sidewise with the similar orbital of the other to form a weak pi bond, which consists of two equal electron clouds distributed above and below the plane of the carbon and hydrogen atoms
- so the carbon-carbon bond consists of one sp2-sp2 sigma bond and one pi bond between the p orbitals not used in the hybridisation, with a bond length of 134 pm, the C-H bond being sp2-s sigma at 108 pm, the H-C-H angle 117.6 degrees and the H-C-C angle 121 degrees

Two sp carbons joined head on for the sigma bond, each also bonded to one hydrogen

Two unhybridised p orbitals on each carbon overlapping sidewise

So the triple bond is one sigma bond and two pi bonds

Figure 4.U Formation of sigma and pi bonds in ethyne

sp in ethyne

- in the formation of ethyne both carbon atoms undergo sp-hybridisation, leaving two unhybridised orbitals, 2py and 2px, on each
- one sp hybrid orbital of one carbon overlaps axially with an sp orbital of the other to form the C-C sigma bond, while the other hybridised orbital of each carbon overlaps axially with the half filled s orbital of a hydrogen atom forming a sigma bond
- each of the two unhybridised p orbitals of both carbons overlaps sidewise to form two pi bonds between the carbon atoms, so the triple bond is made up of one sigma and two pi bonds

4.6.3 Hybridisation of elements involving d orbitals

Which combinations are possible

- the elements present in the third period contain d orbitals in addition to s and p orbitals, and the energies of the 3d orbitals are comparable to those of the 3s and 3p orbitals and also to those of the 4s and 4p orbitals
- as a consequence hybridisation involving either 3s, 3p and 3d, or 3d, 4s and 4p, is possible, however since the difference in energies of the 3p and 4s orbitals is significant, no hybridisation involving 3p, 3d and 4s orbitals is possible

Table 4.D

Shape of molecules/ions	Hybridisation type	Atomic orbitals	Examples
Square planar	dsp2	d+s+p(2)	[Ni(CN)4]2–, [Pt(Cl)4]2–
Trigonal bipyramidal	sp3d	s+p(3)+d	PF5, PCl5
Square pyramidal	sp3d2	s+p(3)+d(2)	BrF5
Octahedral	sp3d2 d2sp3	s+p(3)+d(2) d(2)+s+p(3)	SF6, [CrF6]3– [Co(NH3)6]3+

Five sp3d hybrids pointing to the corners of a trigonal bipyramid

Three lying in one plane at 120 degrees, the equatorial bonds

Two more at right angles to that plane, one above and one below, the axial bonds

Figure 4.V Trigonal bipyramidal geometry of PCl5 molecule

Six sp3d2 hybrids pointing to the corners of a regular octahedron

Each overlapping a singly occupied fluorine orbital

All six S-F bonds equivalent, at 90 degrees to their neighbours

Figure 4.W Octahedral geometry of SF6 molecule

Phosphorus pentachloride, and sulphur hexafluoride

- the five orbitals of phosphorus, one s, three p and one d, hybridise to a set of five sp3d hybrid orbitals directed towards the five corners of a trigonal bipyramid, and not all the bond angles in that geometry are equivalent
- three P-Cl bonds lie in one plane at 120 degrees to each other and are termed equatorial bonds, while the remaining two, one above and one below that plane at 90 degrees to it, are called axial bonds, and since the axial bond pairs suffer more repulsive interaction from the equatorial bond pairs, the axial bonds are slightly longer and hence slightly weaker, which makes PCl5 more reactive
- in SF6 the central sulphur atom has the ground state configuration 3s2 3p4, and in the excited state six orbitals, one s, three p and two d, are singly occupied, hybridising to six sp3d2 orbitals projected towards the six corners of a regular octahedron, each overlapping a singly occupied fluorine orbital to give six S-F sigma bonds

4.7 Molecular orbital theory

The salient features

- molecular orbital theory was developed by F. Hund and R.S. Mulliken in 1932, and in it the electrons in a molecule are present in various molecular orbitals as the electrons of atoms are present in various atomic orbitals, the atomic orbitals of comparable energies and proper symmetry combining to form them
- while an electron in an atomic orbital is influenced by one nucleus, in a molecular orbital it is influenced by two or more nuclei depending on the number of atoms, so an atomic orbital is monocentric while a molecular orbital is polycentric

Two atomic wave functions, one centred on each atom

Adding them builds electron density between the nuclei, the bonding orbital

Subtracting them cancels density between the nuclei, the antibonding orbital

Figure 4.X Formation of bonding (σ) and antibonding (σ*) molecular orbitals by the linear combination of atomic orbitals ψA and ψB centered on two atoms A and B

The rest of the features

- the number of molecular orbitals formed is equal to the number of combining atomic orbitals, so when two atomic orbitals combine, two molecular orbitals are formed, one known as the bonding molecular orbital and the other as the antibonding molecular orbital
- the bonding molecular orbital has lower energy and hence greater stability than the corresponding antibonding one, and just as the electron probability distribution around a nucleus in an atom is given by an atomic orbital, the distribution around a group of nuclei in a molecule is given by a molecular orbital
- molecular orbitals, like atomic orbitals, are filled in accordance with the aufbau principle, obeying Pauli's exclusion principle and Hund's rule

4.7.1 Formation of molecular orbitals, linear combination of atomic orbitals

Adding and subtracting wave functions

- atomic orbitals can be expressed by wave functions representing the amplitude of the electron waves, obtained from the solution of the Schrödinger wave equation, however since it cannot be solved for any system containing more than one electron, molecular orbitals are difficult to obtain directly, so an approximate method known as linear combination of atomic orbitals has been adopted
- applying it to the hydrogen molecule of two atoms A and B, each with one electron in a 1s orbital, the atomic orbitals are represented by the wave functions ψA and ψB, and the molecular orbitals are formed by adding and by subtracting them
- the orbital formed by addition is called the bonding molecular orbital, σ, and the one formed by subtraction is called the antibonding molecular orbital, σ*

Where the electron density sits

- in an antibonding molecular orbital the electron waves cancel each other by destructive interference, so the electron density in a bonding orbital is located between the nuclei of the bonded atoms, which keeps the repulsion between the nuclei very low, while in an antibonding orbital most of the density is away from the space between the nuclei, with a nodal plane on which the density is zero between them, so the internuclear repulsion is high
- electrons in a bonding orbital tend to hold the nuclei together and stabilise the molecule, so a bonding orbital always has lower energy than either of the atomic orbitals that formed it, whereas electrons in the antibonding orbital destabilise the molecule because the mutual repulsion of the electrons there exceeds the attraction to the nuclei
- the energy of the antibonding orbital is raised above that of the parent atomic orbitals and the energy of the bonding orbital lowered below them, the total energy of the two molecular orbitals remaining the same as that of the two original atomic orbitals

4.7.2 Conditions for the combination of atomic orbitals

The three conditions

- the combining atomic orbitals must have the same or nearly the same energy, so a 1s orbital can combine with another 1s orbital but not with a 2s orbital, whose energy is appreciably higher, though this is not true if the atoms are very different
- they must have the same symmetry about the molecular axis, by convention the z-axis, so a 2pz orbital of one atom can combine with the 2pz of the other but not with its 2px or 2py because of their different symmetries
- they must overlap to the maximum extent, since the greater the overlap the greater the electron density between the nuclei

4.7.3 Types of molecular orbitals, and 4.7.4 the energy level diagram

Sigma and pi molecular orbitals

- molecular orbitals of diatomic molecules are designated sigma, pi and delta, the sigma orbitals being symmetrical around the bond axis while the pi orbitals are not
- the linear combination of 1s orbitals centred on two nuclei produces two orbitals symmetrical around the bond axis, designated σ1s and σ*1s, and taking the internuclear axis as z, a combination of 2pz orbitals also produces two sigma orbitals
- orbitals obtained from 2px and 2py are not symmetrical around the bond axis because of positive lobes above and negative lobes below the molecular plane, and are labelled π and π*, a π bonding orbital having larger electron density above and below the internuclear axis while the π* antibonding orbital has a node between the nuclei

Contour maps of each molecular orbital beside the energy level it sits at

From 1s, from 2pz and from 2px in turn, each giving a bonding and an antibonding orbital

The bonding one always below the parent atomic level and the antibonding one above

Figure 4.Y Contours and energies of bonding and antibonding molecular orbitals formed through combinations of (a) 1s atomic orbitals; (b) 2pz atomic orbitals and (c) 2px atomic orbitals.

What this page carries

- the whole of this page is given over to the contours and energies of the bonding and antibonding molecular orbitals
- the diagrams are printed as drawings rather than as text, so they are described here rather than reproduced

The eight molecular orbitals from 2s and 2p

- just as the 1s atomic orbitals on two atoms form σ1s and σ*1s, the 2s and 2p atomic orbitals, eight in all across two atoms, give rise to eight molecular orbitals
- the bonding ones are σ2s, σ2pz, π2px and π2py, and the antibonding ones are σ*2s, σ*2pz, π*2px and π*2py

Two different orders

- the energy levels have been determined experimentally from spectroscopic data for homonuclear diatomic molecules of the second row, and for O2 and F2 the increasing order runs σ1s, σ*1s, σ2s, σ*2s, σ2pz, then the pair π2px and π2py, then the pair π*2px and π*2py, then σ*2pz
- this sequence is not correct for the remaining molecules Li2, Be2, B2, C2 and N2, for which the order runs σ1s, σ*1s, σ2s, σ*2s, then the pair π2px and π2py, then σ2pz, then the pair π*2px and π*2py, then σ*2pz
- the important characteristic feature of this second order is that the energy of the σ2pz molecular orbital is higher than that of the π2px and π2py molecular orbitals

4.7.5 Electronic configuration and molecular behaviour

What the configuration tells you

- the distribution of electrons among the various molecular orbitals is called the electronic configuration of the molecule, and from it a good deal about the molecule can be read off
- writing Nb for the number of electrons occupying bonding orbitals and Na for the number in antibonding orbitals, the stability of the molecule follows from the comparison of the two

Hydrogen, helium, beryllium and lithium

- no unpaired electron is present in the hydrogen molecule, therefore it is diamagnetic
- helium has the configuration 1s2, so He2 would have four electrons filling σ1s and σ*1s, giving a bond order of one half of 2 minus 2, that is 0, so the molecule is unstable and does not exist, and similarly Be2 does not exist
- lithium is 1s2 2s1, so Li2 has six electrons and the configuration σ1s2 σ*1s2 σ2s2, also written KK σ2s2 where KK is the closed K shell, giving four electrons in bonding orbitals and two in antibonding, so the bond order is one half of 4 minus 2, that is 1, and since there are no unpaired electrons Li2 is stable and diamagnetic, as indeed it is found to be in the vapour phase

Carbon, and oxygen

- carbon is 1s2 2s2 2p2, so C2 has twelve electrons and a bond order of one half of 8 minus 4, that is 2, and should be diamagnetic, as diamagnetic C2 molecules detected in the vapour phase confirm, and it is worth noting that the double bond in C2 consists of both pi bonds, because of four electrons in two pi molecular orbitals, whereas in most other molecules a double bond is a sigma bond and a pi bond
- oxygen is 1s2 2s2 2p4, so O2 has sixteen electrons, ten in bonding orbitals and six in antibonding, giving a bond order of one half of 10 minus 6, that is 2, so the atoms are held by a double bond
- O2 also contains two unpaired electrons, in the π*2px and π*2py molecular orbitals, so the molecule should be paramagnetic, a prediction that corresponds to experimental observation, and in this way the theory successfully explains the paramagnetic nature of oxygen

Each molecule from B2 to Ne2 with its molecular orbitals drawn and filled

Below each one its bond energy, bond length, bond order and magnetic behaviour

And the valence electron configuration that goes with it

Figure 4.Z MO occupancy and molecular properties for B2 through Ne2.

4.9 Hydrogen bonding

What a hydrogen bond is

- nitrogen, oxygen and fluorine are highly electronegative elements, and when they are attached to a hydrogen atom by a covalent bond the electrons of that bond are shifted towards the more electronegative atom
- the partially positively charged hydrogen atom then forms a bond with another more electronegative atom, and this bond, known as a hydrogen bond, is weaker than the covalent bond, so in HF the hydrogen bond exists between the hydrogen atom of one molecule and the fluorine atom of another
- the hydrogen bond acts as a bridge between two atoms, holding one by a covalent bond and the other by a hydrogen bond, and is represented by a dotted line while a solid line represents the covalent bond, so a hydrogen bond can be defined as the attractive force which binds the hydrogen atom of one molecule with the electronegative atom, F, O or N, of another

4.9.1 Cause of formation of hydrogen bond

Why the charge separates

- when hydrogen is bonded to a strongly electronegative element X, the electron pair shared between the two atoms moves far away from the hydrogen atom, so the hydrogen becomes highly electropositive with respect to X
- since there is a displacement of electrons towards X, the hydrogen acquires a fractional positive charge while X attains a fractional negative charge, which results in a polar molecule with an electrostatic force of attraction between neighbours
- the magnitude of hydrogen bonding depends on the physical state of the compound, being maximum in the solid state and minimum in the gaseous state, so hydrogen bonds have a strong influence on the structure and properties of compounds

The hydroxyl hydrogen of o-nitrophenol lying between two oxygen atoms

Both oxygens belonging to the same molecule

So the bond closes a ring within the molecule rather than joining two of them

Figure 4.AA Intramolecular hydrogen bonding in o-nitrophenol molecule

4.9.2 Types of H-bonds

Between molecules, and within one

- an intermolecular hydrogen bond is formed between two different molecules of the same or different compounds, as in HF, alcohol or water molecules
- an intramolecular hydrogen bond is formed when a hydrogen atom lies between two highly electronegative atoms, F, O or N, present within the same molecule, as in o-nitrophenol where the hydrogen is between the two oxygen atoms

Summary

From Kössel and Lewis to the crystal lattice

- Kössel's first insight into the mechanism of formation of electropositive and electronegative ions related the process to the attainment of noble gas configurations by the respective ions, electrostatic attraction between ions being the cause of their stability, which gives the concept of electrovalency
- the first description of covalent bonding was provided by Lewis in terms of the sharing of electron pairs between atoms, related to the attainment of noble gas configurations by the reacting atoms, the Lewis dot symbols showing the number of valence electrons and the Lewis dot structures showing pictorial representations of bonding in molecules
- an ionic compound is pictured as a three-dimensional aggregation of positive and negative ions in an ordered arrangement called the crystal lattice, in which there is a charge balance between the positive and negative ions, the lattice being stabilized by the enthalpy of lattice formation
- while a single covalent bond is formed by sharing an electron pair, multiple bonds result from sharing two or three pairs, and some bonded atoms have additional pairs not involved in bonding, called lone pairs, while bond length, bond angle, bond enthalpy, bond order and bond polarity all have a significant effect on the properties of compounds

Resonance, VSEPR and valence bond

- a number of molecules and polyatomic ions cannot be described accurately by a single Lewis structure, so several descriptions based on the same skeletal structure are written and taken together represent the molecule or ion, which is the concept of resonance, the contributing structures or canonical forms together constituting the resonance hybrid
- the VSEPR model used for predicting geometrical shapes is based on the assumption that electron pairs repel each other and therefore tend to remain as far apart as possible, so molecular geometry is determined by repulsions in the order lone pair against lone pair, then lone pair against bond pair, then bond pair against bond pair
- the valence bond approach is concerned with the energetics of covalent bond formation, about which the Lewis and VSEPR models are silent, discussing bond formation in terms of overlap of orbitals, so that in H2 the potential energy of the system falls as the two atoms approach and touches a minimum at the bond distance, any attempt to bring the nuclei closer raising the energy sharply, though the actual bond enthalpy and bond length are not obtained by overlap alone

Hybridisation, molecular orbitals and the hydrogen bond

- for explaining the characteristic shapes of polyatomic molecules Pauling introduced hybridisation of atomic orbitals, sp, sp2 and sp3 hybridisations of Be, B, C, N and O explaining the formation and shapes of BeCl2, BCl3, CH4, NH3 and H2O, and also the multiple bonds in C2H2 and C2H4
- molecular orbital theory describes bonding in terms of the combination and arrangment of atomic orbitals to form molecular orbitals associated with the molecule as a whole, their number always equal to the number of atomic orbitals from which they are formed, bonding orbitals increasing electron density between the nuclei and lying lower in energy than the individual atomic orbitals while antibonding orbitals have a region of zero density between the nuclei and more energy
- the electronic configuration of a molecule is written by filling electrons in the molecular orbitals in order of increasing energy, the Pauli exclusion principle and Hund's rule applying as they do for atoms, and molecules are stable if the number of electrons in bonding orbitals is greater than that in antibonding orbitals
- a hydrogen bond is formed when a hydrogen atom finds itself between two highly electronegative atoms such as F, O and N, and may be intermolecular or intramolecular, hydrogen bonds having a powerful effect on the structure and properties of many compounds

Exercises

Exercises 4.1 to 4.3

- 4.1 Explain the formation of a chemical bond.
- 4.2 Write Lewis dot symbols for atoms of the following elements : Mg, Na, B, O, N, Br.
- 4.3 Write Lewis symbols for the following atoms and ions: S and S2–; Al and Al3+; H and H–
- 4.4 Draw the Lewis structures for the following molecules and ions : H2S, SiCl4, BeF2, CO32–, HCOOH
- 4.5 Define octet rule. Write its significance and limitations.

Exercises, continued

Exercises 4.6 to 4.15

- 4.6 Write the favourable factors for the formation of ionic bond.
- 4.7 Discuss the shape of the following molecules using the VSEPR model: $BeCl_2$, BCl_3, $SiCl_4$, AsF_5, H_2S, PH_3
- 4.8 Although geometries of NH_3 and H_2O molecules are distorted tetrahedral, bond angle in water is less than that of ammonia. Discuss.
- 4.9 How do you express the bond strength in terms of bond order ?
- 4.10 Define the bond length.
- 4.11 Explain the important aspects of resonance with reference to the CO_3^{2-} ion.
- 4.12 H_3PO_3 can be represented by structures 1 and 2 shown below. Can these two structures be taken as the canonical forms of the resonance hybrid representing H_3PO_3 ? If not, give reasons for the same.
- 4.13 Write the resonance structures for SO_3, NO_2 and NO_3^-.
- 4.14 Use Lewis symbols to show electron transfer between the following atoms to form cations and anions : (a) K and S (b) Ca and O (c) Al and N.
- 4.15 Although both CO_2 and H_2O are triatomic molecules, the shape of H_2O molecule is bent while that of CO_2 is linear. Explain this on the basis of dipole moment.

Exercises 4.16 to 4.25

- 4.16 Write the significance/applications of dipole moment.
- 4.17 Define electronegativity. How does it differ from electron gain enthalpy ?
- 4.18 Explain with the help of suitable example polar covalent bond.
- 4.19 Arrange the bonds in order of increasing ionic character in the molecules: LiF, K_2O, N_2, SO_2 and ClF_3.
- 4.20 The skeletal structure of CH_3COOH as shown below is correct, but some of the bonds are shown incorrectly. Write the correct Lewis structure for acetic acid.
- 4.21 Apart from tetrahedral geometry, another possible geometry for CH_4 is square planar with the four H atoms at the corners of the square and the C atom at its centre. Explain why CH_4 is not square planar ?
- 4.22 Explain why BeH_2 molecule has a zero dipole moment although the Be–H bonds are polar.
- 4.23 Which out of NH_3 and NF_3 has higher dipole moment and why ?
- 4.24 What is meant by hybridisation of atomic orbitals? Describe the shapes of sp, sp^2, sp^3 hybrid orbitals.
- 4.25 Describe the change in hybridisation (if any) of the Al atom in $AlCl_3$ combining with a chloride ion to give the tetrachloroaluminate ion.

Exercises, concluded

Exercises 4.26 to 4.33

- 4.26 Is there any change in the hybridisation of B and N atoms as a result of the following reaction?
- 4.27 Draw diagrams showing the formation of a double bond and a triple bond between carbon atoms in C2H4 and C2H2 molecules.
- 4.28 What is the total number of sigma and pi bonds in the following molecules? (a) C2H2 (b) C2H4
- 4.29 Considering x-axis as the internuclear axis which out of the following will not form a sigma bond and why? (a) 1s and 1s (b) 1s and 2px; (c) 2py and 2py (d) 1s and 2s.
- 4.30 Which hybrid orbitals are used by carbon atoms in the following molecules? CH3–CH3; (b) CH3–CH=CH2; (c) CH3-CH2-OH; (d) CH3-CHO (e) CH3COOH
- 4.31 What do you understand by bond pairs and lone pairs of electrons? Illustrate by giving one exmaple of each type.
- 4.32 Distinguish between a sigma and a pi bond.
- 4.33 Explain the formation of H2 molecule on the basis of valence bond theory.

Exercises 4.34 to 4.40

- 4.34 Write the important conditions required for the linear combination of atomic orbitals to form molecular orbitals.
- 4.35 Use molecular orbital theory to explain why the Be2 molecule does not exist.
- 4.36 Compare the relative stability of the following species and indicate their magnetic properties: the superoxide ion and the peroxide ion.
- 4.37 Write the significance of a plus and a minus sign shown in representing the orbitals.
- 4.38 Describe the hybridisation in case of PCl5. Why are the axial bonds longer as compared to equatorial bonds?
- 4.39 Define hydrogen bond. Is it weaker or stronger than the van der Waals forces?
- 4.40 What is meant by the term bond order? Calculate the bond order of : N2, O2, O2+ and O2–.